

SPH4U Unit 2 Energy

Chapter 4 Work and Energy Lesson Package

Schoolwork	Useful Links/ Videos/ Handouts	Homework
4.1 Work Done by a Constant Force		
1. Lesson 4.1 Presentation 2. Lesson 4.1 Lesson Package	1. Work and Energy 2. An introduction of how work is changed into kinetic energy: car being pushed 3. Work changed to kinetic energy and the work to stop the object: Ball hits your head 4. Doing work to make something go faster and how to calculate how fast it will be moving: Pushing a bed that's already moving	1. 4.1 Check Your Understanding 2. Textbook pg. 170 #1-8 3. TIPERs pg. 149 B4-RT08; pg 156 - 157 B4-LMCT18, B4-SCT19;
4.2 Kinetic Energy and the Work-Energy Theorem		
1. Lesson 4.2 Presentation 2. Lesson 4.2 Lesson Package	Videos: 1. Kinetic Energy and Potential Energy	1. 4.2 Check Your Understanding 2. Textbook pg. 176 #1-11 3. TIPERs pg 147 – 148 B4-RT05, B4-RT06, B4-RT07, pg 151 B4-RT10; pg 154-155 B4-BCT15, B4-BCT16, B4-BCT17; pg 159 B4-QRT22; pg 160 B4-SCT24; pg 165 B4-BCT30
4.3 Gravitational Potential Energy		
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4.5 The Law of Conservation of Energy		
1. Lesson 4.5 Presentation 2. Lesson 4.5 Lesson Package	Videos: 1. Conservation of Energy: Free Fall, Springs, and Pendulums 2. Converting from gravitational energy to kinetic energy in a roller coaster	1. 4.5 Check Your Understanding 2. Textbook pg. 191 #1-8 3. TIPERs pg 161 B4-RT25; pg 162 B4-RT26; B4-CT27; pg 169 B4-BCT34; pg 171 B4-BCT36; pg. 174-177 B4-WBT39, B4-WBT40, B4-WWT41, B4-WWT42, B4-QRT43,

	3. An example of gravitational energy to kinetic energy on a slide 4. Complex roller coaster problem. Find the height needed for a loop	B4-QRT44; pg 179 B4-RT46; pg 180 B4-CT47, pg. 182 B4-CT49
4.6 Elastic Potential Energy and Simple Harmonic Motion		
1. Lesson 4.6 Presentation 2. Lesson 4.6 Lesson Package	Videos: 1. Hooke's Law Introduction - Force of a Spring 2. Simple Harmonic Motion 3. Elastic energy stored in a spring is converted to kinetic energy 4. Elastic energy stored in a spring is converted to kinetic energy in the vertical direction 5. Another method to solve an elastic energy stored in a spring converted to kinetic energy in the vertical direction	1. 4.6 Check Your Understanding 2. Textbook pg. 200 #1-10 3. TIPERs pg 170 B4-BCT35; pg. 172 B4-BCT37; pg. 173 B4-BCT38
4.7 Springs and Conservation of Energy		
1. Lesson 4.7 Presentation 2. Lesson 4.7 Lesson Package		1. Textbook pg. 208 #1-10
Chapter 4 Review		
1. Chapter 4 Self- Quiz pg. 213 (attempt all) 2. Chapter 4 Review pg. 214 #1-63, 67, 68 3. Communication Questions (lesson package)		1. Chapter 4 Self- Quiz pg. 213 (attempt all) 2. Chapter 4 Review pg. 214 #1-63, 67, 68 3. Chapter 4 - Check Your Knowledge – Definitions (lesson package) 4. Communication Questions (lesson package) 5. Check Atomi videos and questions

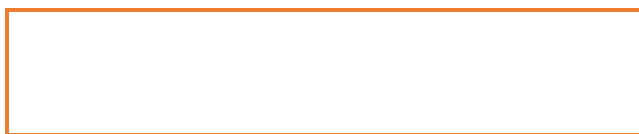
Lesson 4.1 Work Done by a Constant Force

Lesson Goal: Explain the concept of work and describe what conditions must be met for work to be done on an object

Success Criteria: At the end of this lesson, I will be able to:

- ✓ Define what work is.
- ✓ Describe what conditions must be met for work to be done on an object
- ✓ Understand when work done on an object is zero, positive, or negative
- ✓ Calculate work done by a single force
- ✓ Calculate work done by multiple forces
- ✓ To find work from the area under a graph of the force exerted along the direction of motion versus the position of the point at which that force is exerted

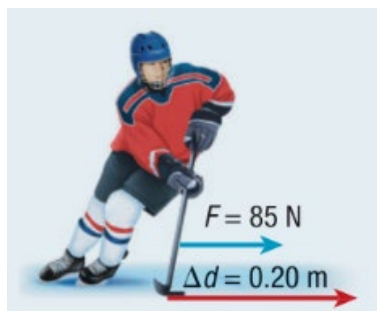
(Mechanical) Work, W , is the product of the magnitude of an object's displacement and the component of the applied force in the direction of the displacement, given by the following equation:



Example 1: Calculating the Work Done When the Force Is Parallel to the Displacement

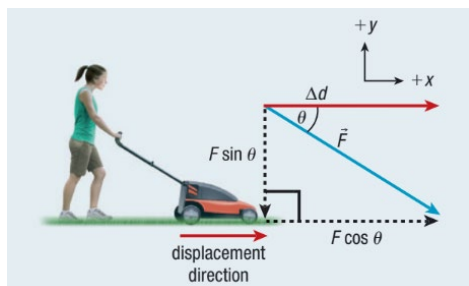
The displacement of an object is often in the same direction as the applied force. In this case, the equation for calculating the work done by the force is simplified.

Suppose a hockey player slides a puck along the ice with a constant force of 85 N in the forward direction (figure below). The puck moves a horizontal distance of 0.20 m while in contact with the hockey stick. Calculate the amount of work done on the puck by the stick.



Example 2: Calculating the Work Done When the Force Is at an Angle to the Displacement

The applied force and the displacement can be at an angle to each other. For example, a student pushes a lawnmower forward with a constant force of 48 N for a distance of 7.5 m (figure below). The angle between the force and the displacement of the lawnmower is 32° . Calculate how much work is done on the lawnmower by the student.



Example 3: Calculating the Work Done When the Force Is Perpendicular to the Displacement

Suppose you carry books with a weight of 22 N a distance of 3.8 m across a room (figure below). Determine the work done on the books.

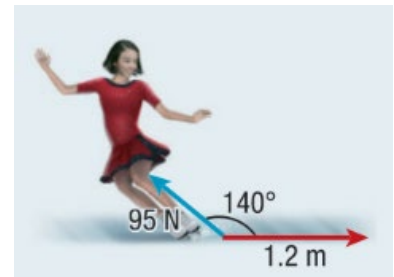


Example 4: Calculating Negative Work in One Dimension

Suppose a car is moving along a straight road when the driver suddenly applies the brakes. The force of friction between the ground and the car tires is opposite to the car's direction of motion and decreases the car's speed. Calculate the work done by a constant frictional force of 1.4 kN over a distance of $1.2 \times 10^2 \text{ m}$.

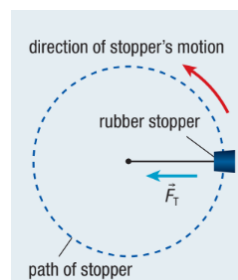
Example 5: Calculating Negative Work in Two Dimensions

An ice skater slides to a stop by pushing her blades against the ice (figure below). The ice exerts a constant force of 95 N on the skater, and the skater stops in 1.2 m . The angle between the force and the skater's direction of motion is 140° . Calculate the work done on the skater by the ice.



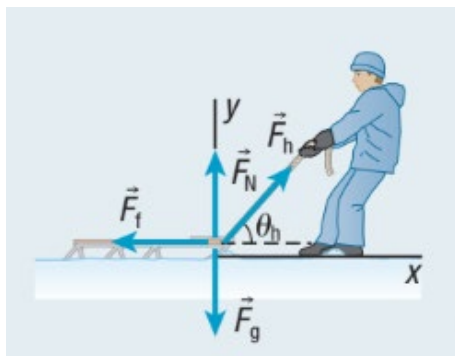
Example 6: Calculating the Work Done on an Object Moving in a Circular Path

A student attaches one end of a string to a rubber stopper. She then holds the other end of the string and twirls the rubber stopper in a horizontal circle around her head. The string exerts a tension force on the stopper directed toward the centre of the circle (figure below). Determine the work done on the stopper by the string during a revolution.



Example 7: Calculating the Work Done by Multiple Forces on a Dragged Object

The hiker exerts a constant force of 135 N on the sled at a 48.0° angle to the sled's displacement. At the same time, a constant 67.0 N frictional force on the sled from the snow opposes the motion. The sled also experiences the force from gravity and the normal force from the snow, but these forces do not contribute to the work. Calculate the work done by the hiker (W_h), the work done by friction (W_f), and the total (W_T) work done on the sled when the hiker pulls the sled 345 m over the snow.



Summary of Positive, Negative and Zero Work

Force versus displacement graph

The work done by a force on an object may be represented graphically by plotting the magnitude of the force, F , on the y-axis and the magnitude of the object's position, d , on the x-axis. In such cases, we assume that the force acts in the same direction as the object's displacement or in a direction opposite the object's displacement. Figure 7 shows a force-position (F - d) graph for a **constant force** acting through a displacement d .

The work done, W , is equal to the area under the F - d graph (the shaded rectangle) and is equal in value to the product $F \cdot d$.

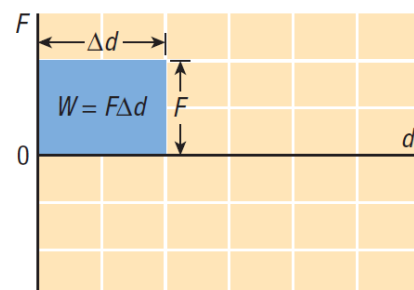


Figure 7 F - d graph for a constant force acting through a displacement

Positive and negative work may also be represented using an F - d graph. In Figure 8, positive work (work done when force and displacement are in the same direction) is represented by the blue rectangle above the d -axis. Negative work (work done when force and displacement are in opposite directions) is represented by the red rectangle below the d -axis.

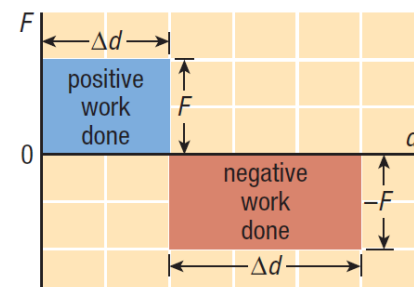
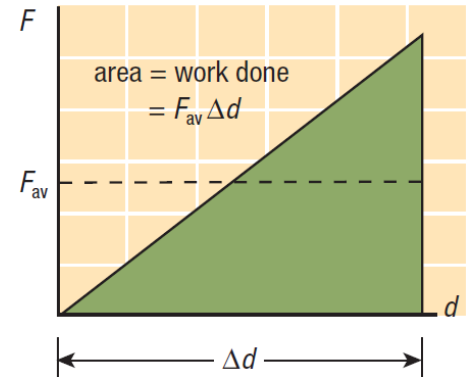


Figure 8 F - d graph representing positive and negative work done

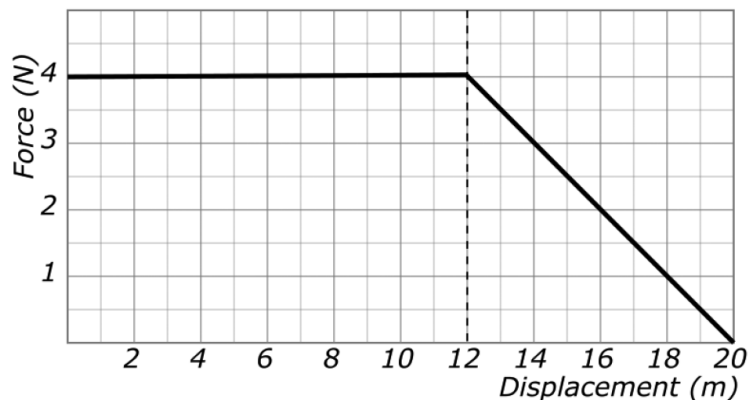
Work Done by a Changing Force

The equation $W = F \cdot d$ or $W = F(\cos\theta) \cdot d$ can be used only to calculate the mechanical work done on an object when the force on the object is constant. However, in many cases, a force varies in magnitude during a displacement. For example, when a bus driver steadily depresses the accelerator pedal of a bus, the force the engine applied on the wheels increases uniformly. The force is not constant.

The work done on an object by a changing force that is in the same direction as the object's displacement may be represented using an F - d graph. Figure 9 shows an F - d graph for a uniformly increasing force. As in the case of a constant force, the work done is equal to the area under the F - d graph. In this case, the work done, W , is equal in value to the product $F_{av} \cdot d$. F_{av} represents the average force applied to the object as it is displaced.



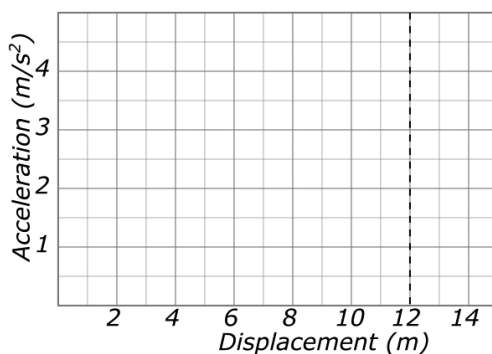
Scenario 1: Starting from rest, a 1 kg cart moves to the right along a horizontal surface that exerts negligible friction, while being pushed by a horizontal force F as shown. The force exerted on the cart as a function of displacement is graphed below.



a) Is the claim correct: The cart will have a change in mechanical energy of 48 J as its displacement is changed by 12 m.

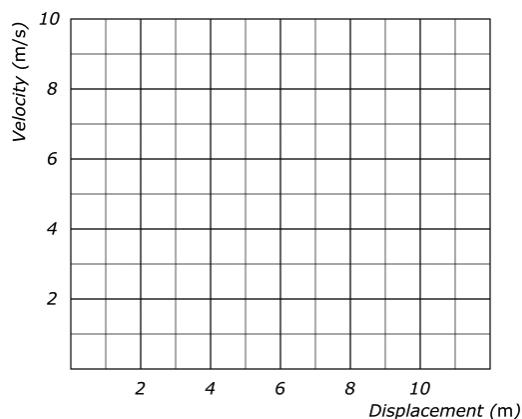
b) Explain how the graph above could be used to determine the final speed of the cart after 12 m.

c) On the grid below, sketch a graph of the acceleration of the cart as a function of displacement from $x = 0$ m to $x = 12$ m.



d) After creating her graph of acceleration vs. displacement, Angela says, “Oh, the slope of a velocity graph represents the acceleration. So, since the acceleration is constant at 4 m/s^2 , the velocity should be a line with a slope of 4 m/s^2 . Do you agree with Angela? Explain.

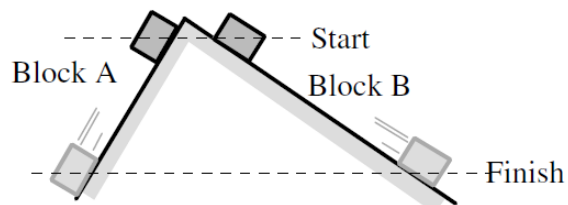
e) Sketch a graph of the velocity of the cart as a function of displacement.



f) The equation $v_f^2 = v_i^2 + 2a(\Delta x)$ can help describe the velocity of an object with a constant acceleration as a function of position. Is this equation consistent with the graph you sketched in part e? Explain.

TIPERs B4-SCT20: BLOCKS SLIDING DOWN FRICTIONLESS RAMPS—WORK BY THE EARTH

Two identical blocks are released from rest at the same height. Block A slides down a steeper ramp than Block B. Both ramps are frictionless. The blocks reach the same final height indicated by the lower dashed line. Three students are comparing the work done on the two blocks by the gravitational force (the weight of the blocks):



Asmita: “Work is related to the product of force and displacement, and the weight is the same since the blocks are identical. But Block B travels farther, so more work is done on Block B by the gravitational force than on Block A.”

Ben: “Both blocks fall the same vertical distance, so the work done is the same.”

Cocheta: “By Newton’s third law, the force exerted on the block by Earth is exactly cancelled by the force exerted on Earth by the block. The work done is zero.”

Danae: “The work depends on the angle that the force makes with the displacement. If we put the displacement and force vectors tail-to-tail, the angle is smaller for Block B than for Block A, and so the work done is greater.”

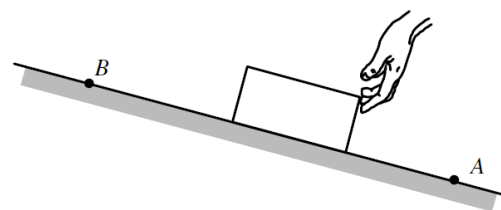
With which, if any, of these students do you agree?

Asmita _____ *Ben* _____ *Cocheta* _____ *Danae* _____ None of them _____

Explain your reasoning.

TIPERs B4-QRT21: BLOCK ON RAMP WITH FRICTION—WORK

A block is pushed at constant speed up a ramp from point A to point B. The direction of the force on the block by the hand is horizontal. There is friction between the block and the ramp. The distance between points A and B is 1m.



For each question below, choose one of the following options and then explain your choice:

(i) is *zero*.

(ii) is *negative*.

(iii) is *positive*.

(iv) *could be positive or negative* depending on the choice of coordinate systems.

a) The work done on the block by the hand as the block travels from point *A* to point *B* _____

Explain your reasoning.

b) The work done on the block by the normal force from the ramp as the block travels from point *A* to point *B* _____

Explain your reasoning.

c) The work done on the block by the friction force from the ramp as the block travels from point *A* to point *B* _____

Explain your reasoning.

d) The work done on the block by the gravitational force of the earth as the block travels from point *A* to point *B* _____

Explain your reasoning.

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding - Lesson 4.1

1. Give two definitions of work.
2. Provide two examples to demonstrate that not every action produces positive work.
3. Describe the most effective way to apply force to push a cart of books across the hall. What do you need to consider making the process more effective?
4. How can a force vs. position graph can be used to determine work done.

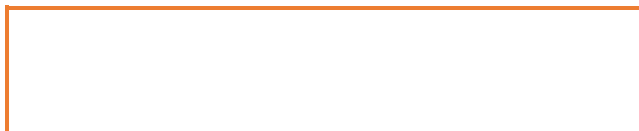
4.2 Kinetic Energy and the Work-Energy Theorem

Lesson Goal: Define kinetic energy and explain the relationship between work and kinetic energy for an object.

Success Criteria: At the end of this lesson, I will be able to:

- ✓ Define kinetic energy
- ✓ Understand that kinetic energy is a scalar quantity
- ✓ Calculate the kinetic energy of an object
- ✓ Explain the relationship between work and kinetic energy for an object.
- ✓ Understand that the work-energy theorem is valid whether the path is curved or straight and whether the forces are constant or varying.
- ✓ Use the work-energy theorem to find the initial or final speed of an object
- ✓ Know how to draw Work-Energy Bar Diagrams

Kinetic Energy, E_k , is the energy an object has due to its motion. An object's kinetic energy is directly proportional to its mass and the square of its speed, according to the following equation:



Derive the units of kinetic energy. What do you notice

Example 1: Calculating Kinetic Energy

A car has a mass of 1.50×10^3 kg and is travelling at a speed of 85.0 km/h. Calculate the car's kinetic energy.

Example 2: Using Kinetic Energy to Determine the Speed of an Object

When fleeing a predator, a 1.4 kg rabbit has a kinetic energy of 96 J. Calculate the speed of the rabbit.

Kinetic Energy and the Work-Energy Theorem

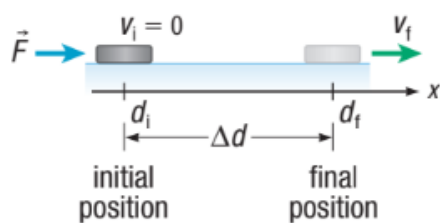


Figure 2 When a force, F , acts on this hockey puck, the puck accelerates. The force does work on the puck, and the puck's kinetic energy changes.

Work-Energy Theorem – the total work done on an object equals the change in its kinetic energy.

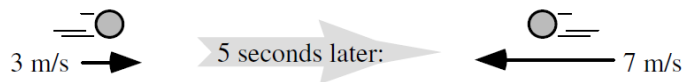
$$W = \Delta E_k$$

Note: The work – energy theorem is only true if no energy losses occur.

TIPERs (B4-WWT03): OBJECT CHANGING VELOCITY—WORK

A 2-kg object accelerates as a net force acts on it. During the 5 seconds this force acts, the object changes its velocity from 3 m/s east to 7 m/s west.

A student states:

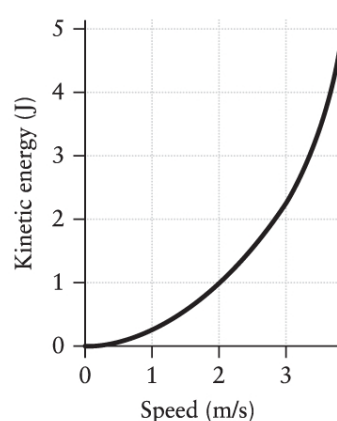


“The initial kinetic energy of the object was 9 Joules, and the final kinetic energy was 49 Joules. Thus the change in kinetic energy of this object during these 5 seconds was 40 J, and thus the work done on this object by the net force during this period was also 40 J.”

What, if anything, is wrong with this statement? If something is wrong, identify it and explain how to correct it. If this statement is correct, explain why.

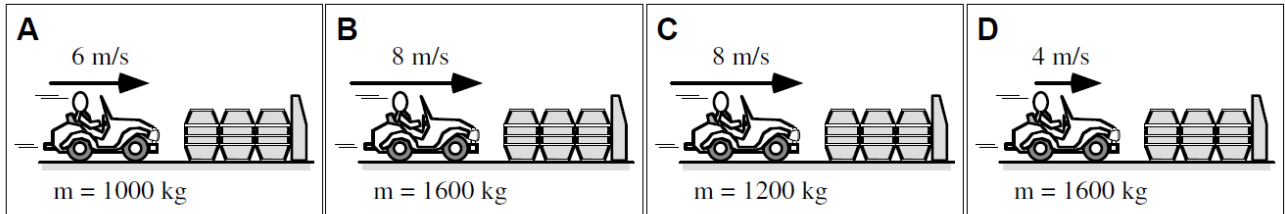
Problem 1 (BFW): A single constant force is exerted on an object. The graph describes the object’s energy.

How much work was done to increase the speed of the object from 1.0 m/s to 3.0 m/s?



TIPERs B4-RT04: CARS AND BARRIERS—STOPPING FORCE IN SAME DISTANCE

Cars that are moving along horizontal roads are going to be stopped by plowing into barrel barriers. All of the cars are the same size and shape, but they are carrying loads with different masses. All of the cars are going to be stopped in the same distance.



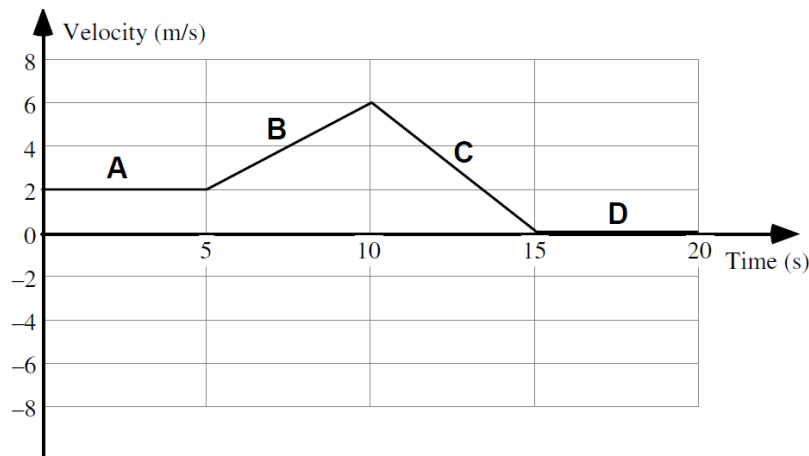
Rank the strength (magnitude) of the forces that will be needed to stop the cars in the same distance.

				OR			
1	2	3	4		All	All	Cannot
Greatest			Least		the same	zero	determine

Explain your reasoning.

TIPERs B4-RT09: VELOCITY-TIME GRAPH I—WORK DONE ON BOX

Shown below is a graph of velocity versus time for an object that moves along a straight, horizontal line under the perhaps intermittent action of a single force exerted by an external agent.



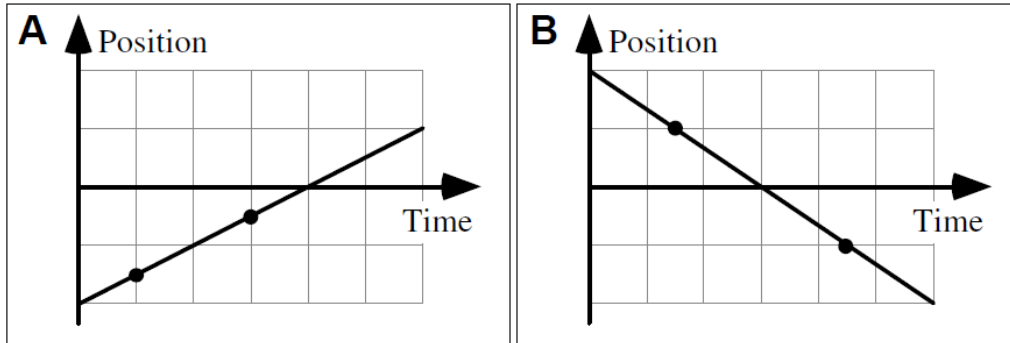
Rank the work done on the box by the external agent for the 5-second intervals shown on the graph.

				OR			
1	2	3	4		All	All	Cannot
Greatest			Least		the same	zero	determine

Explain your reasoning.

TIPERs B4-WWT13: BOAT POSITION-TIME GRAPHS—WORK

Shown are graphs of the position versus time for two boats traveling along a narrow channel. The scales on both axes are the same for the graphs. In each graph, two points are marked with dots.



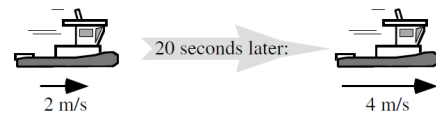
A student who is using these graphs to compare the net work done on the two boats between the two points says:

"I think that more net work was done on the boat in graph B because it moved farther during the interval between the points."

What, if anything, is wrong with this statement? If something is wrong, identify it and explain how to correct it. If this statement is correct, explain why.

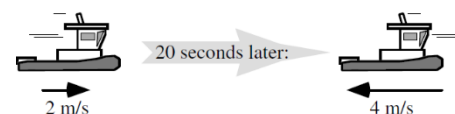
TIPERs B4-BCT14: TUGBOAT CHANGING VELOCITY I—WORK & KINETIC ENERGY BAR CHART

(a) The velocity of a tugboat increases from 2 m/s to 4 m/s in the same direction while a force is applied to the tugboat for 20 seconds.



Draw a work and kinetic energy bar chart for this process. The bar heights should be in correct proportion to one another.

b) The velocity of a tugboat changes from 2 m/s to 4 m/s in the opposite direction while a force is applied to the tugboat for 20 seconds.

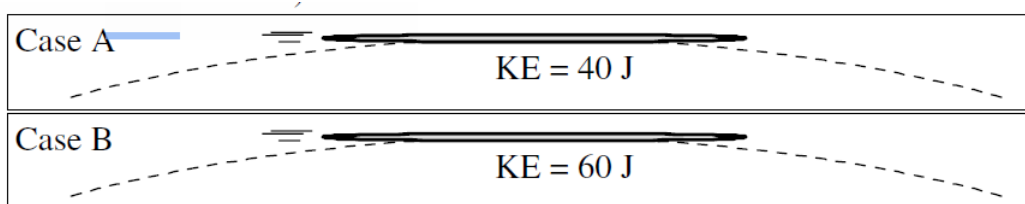


Draw a work and kinetic energy bar chart for this process. The bar heights should be in correct proportion to one another.

Explain.

TIPERs B4-CT23: THROWN JAVELINS—HORIZONTAL FORCE

Shown are two javelins (light spears) that have been thrown at targets. We are viewing the javelins when they are in the air about halfway to landing. Both javelins have the same mass, but they have different kinetic energies as shown. (Ignore air resistance for this task.)



Is the horizontal force acting on the javelin in Case A (i) *greater than*, (ii) *less than*, or (iii) *equal to* the horizontal force acting on the javelin in Case B? _____

Explain your reasoning.

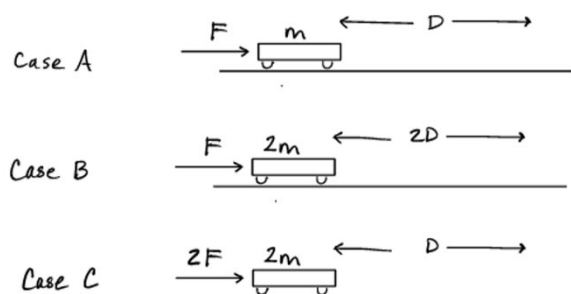
Example 3: Using the Work–Energy Theorem to Calculate Work Done

A blue whale with a mass of 1.5×10^5 kg is swimming with a speed of 6.1 m/s. A nearby boat startles the whale, and the whale increases its speed to 12.8 m/s. Calculate the work done on the whale by the water.

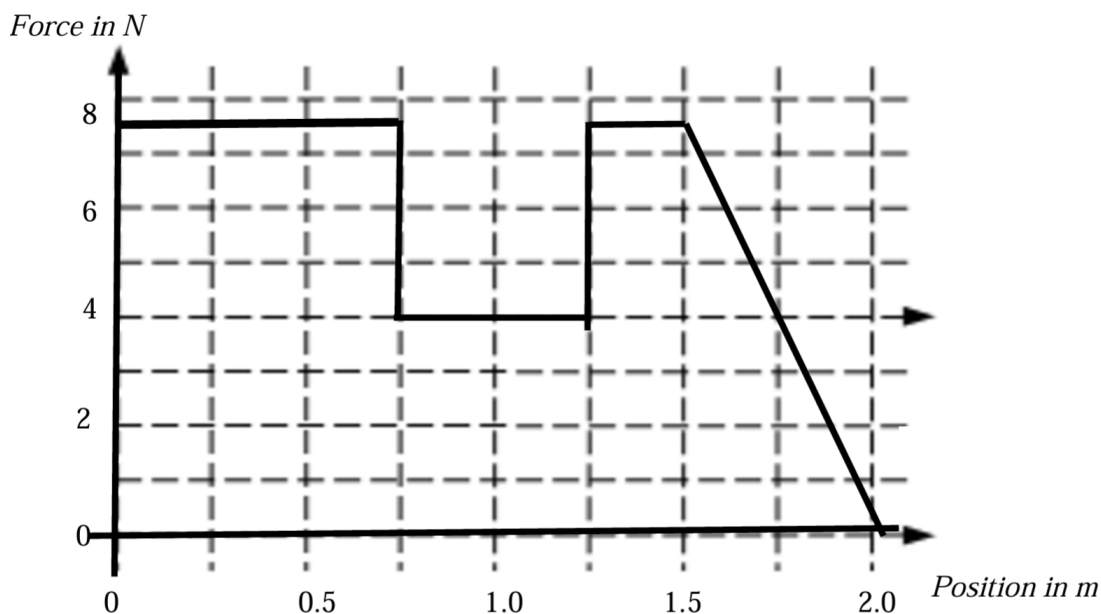
Example 4: Applying the Work–Energy Theorem in the Presence of Friction

A shuffleboard player wants to slide a 430 g disc a distance of precisely 12 m. If the coefficient of kinetic friction between the disc and the playing surface is 0.62, calculate the initial speed at which the player must release the disc.

Problem 1 (BFW) Several carts are being pushed along a horizontal surface by a constant, external force exerted on them as shown in the figure below. The magnitude of the force, the mass of the cart, and the distance pushed are given in each case. Friction between the cart and the surface can be considered to be negligible. Which of the following is the correct ranking of the cases in terms of the kinetic energy of the cart after being pushed the given distance?



Problem 2 (BFW): F vs x graph



Some students take a force probe and pull a cart while recording the tension in a string as a function of time to obtain the graph shown above.

a) i. Explain how the energy of the cart is changing from 0 to 0.75 m without using equations.

ii. Explain how the motion of the cart is changing from 0.75 to 1.25 m as compared with the previous time interval without using equations.

iii. What does the slope represent between 1.5 m and 2.0 m?

The students know the cart started at v_0 , the mass of the object on the end of the rope is m , and the cart is on a surface, pulled only by the string.

b) i. Find the work done by the tension force between 0 and 1.5 m.

ii. Find the speed of the cart at 1.5 m. Express your answer in terms of v_0 , W , m , and fundamental constants.

c) Is the cart moving faster at 1.25 m or at 2.0 m?

Faster at 1.25 m _____ Faster at 2.0 m _____ The speeds are equal _____

Justify your answer.

Problem 3 (BFW): Reasoning in Physics – Forces and Kinetic Energy

A constant force F_0 pushes two objects, one of mass m and the other of mass $4m$, for a total time Δt each, starting from rest. Which of the two objects, the larger or the smaller, has a greater kinetic energy at the end of that time Δt ? Justify your choice using the evidence-reasoning-conclusion framework outlined below.

Evidence: (Represent this situation using graphs, equations, free-body diagrams, etc.)

Reasoning: (Explain how these representations lead to a conclusion.)

Conclusion: (State your choice as a clear answer to the question.)

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding – Lesson 4.2

1. Explain what does a positive and a negative difference mean when calculating the change in kinetic energy.
2. If positive work is done on a system, and only the kinetic energy of the system has changed, how is the velocity of the system affected?
3. State the kinetic energy equation in words.
4. Kinetic energy of an object is directly proportional to the square of the velocity of the object, so the energy required to accelerate an object that is moving is much greater than the energy required to accelerate an object at rest. What is a way to reduce the energy required to accelerate a moving object? State an example.
5. What is 1 J in terms of SI base units?
6. In a car collision, is the work done on a given car equal to the change in kinetic energy?
7. When we derived the work-energy theorem, we assumed that the force is constant. Would the theorem hold true if the applied force is not constant?

4.3 Gravitational Potential Energy

Lesson Goal: Define gravitational potential energy and mechanical energy and solve problems related to them.

Success Criteria: At the end of this lesson, I will be able to:

- ✓ Define gravitational potential energy
- ✓ Define mechanical energy
- ✓ Understand that potential energy is associated with the configuration of a system. , unlike kinetic energy, which is associated with a property of a single object (its speed)
- ✓ Gravitational potential energy near the surface of Earth increases with separation of the object and Earth, measured in terms of the height of the object relative to the surface of Earth
- ✓ Outline the process of calculating apparent weight

Gravitational Potential Energy, E_g , is the stored energy an object has due to its position and the applied gravitational force. An object's gravitational energy is directly proportional to its mass, acceleration due to gravity, and the object's change in height as measured from a reference height.



Note: In the above formula, g is a scalar b/c we are only interested in the magnitude of the acceleration. The sign of the change in E_g comes from the change in Δy .

Analyzing gravitational potential energy mathematically

Units of Gravitational Potential Energy

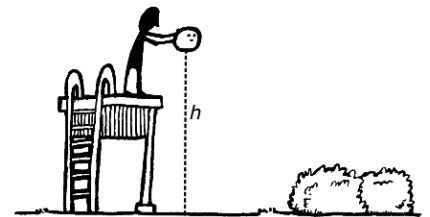
Problem 1(BFW) A ball is dropped from the top of an eleven-story building to a balcony on the ninth floor. In which case is the change in the potential energy associated with the motion of the ball the greatest?

- a) It is greatest when $y=0$ is set at ground level.
- b) It is greatest when $y=0$ is set level with the ninth-floor balcony.
- c) It is greatest when $y=0$ is set level with the top of the building.
- d) The change in potential energy is the same in all three cases.

Justify your answer.

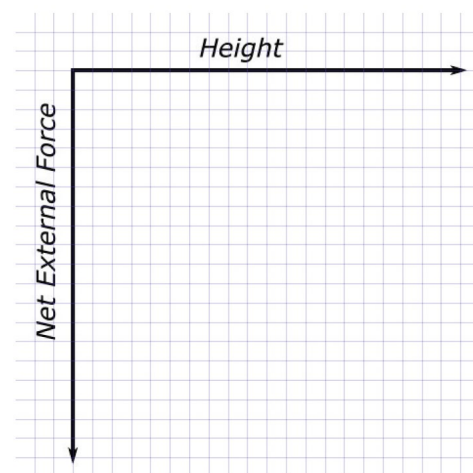
Scenario 1: A ball falls from rest from a height h . Consider the ball to be the system.

- a) Draw a dotted circle around the system.

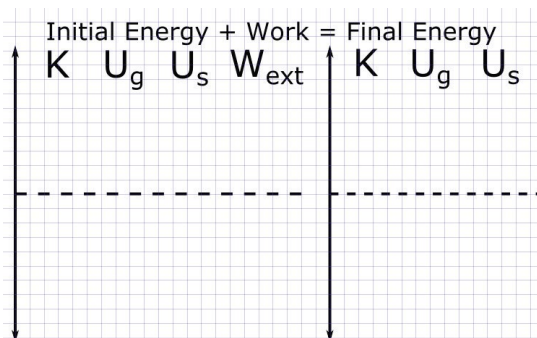


- b) Draw a free-body diagram showing and labeling the forces (not components) exerted on the ball. Represent the ball by a dot. (You should only include forces exerted on the system.) Draw the relative lengths of all vectors to reflect the relative magnitudes of all the forces. Each force must be represented by a distinct arrow starting on and pointing away from the dot.

c) On the grid on the right, sketch a graph of the net external force on the ball vs. height. What are the units of the area under the curve? What does the area under the curve represent?

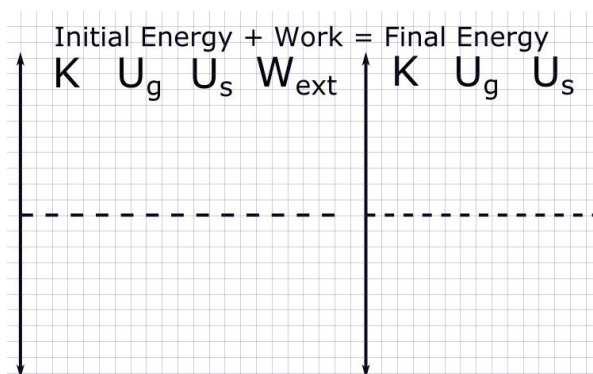


d) Fill in the energy bar chart to describe the energy of the system from when the ball is dropped to just before it hits the ground. Explain your reasoning.



e) Does the mechanical energy increase, decrease, or stay the same as the ball falls from height h ? Justify your answer by referencing the energy bar chart created in part d).

f) If the system were to include Earth, how would that change the energy bar chart? Sketch a new chart and discuss the changes.



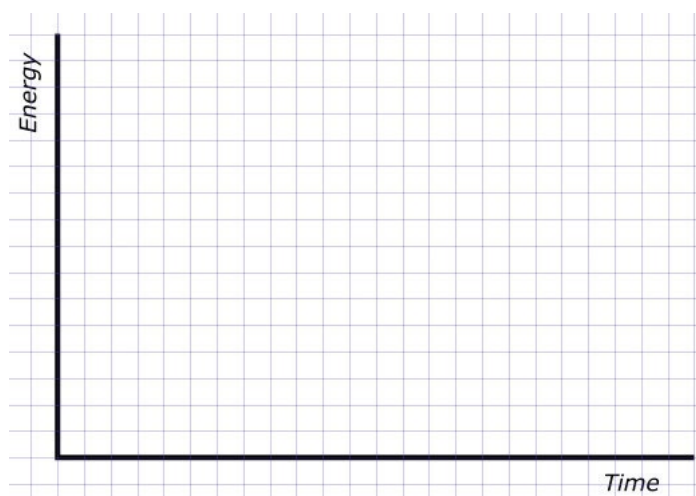
g) If the system were to include Earth, would the mechanical energy increase, decrease, or stay the same while the ball was falling? Justify your answer by referencing the energy bar chart created in Part F.

Scenario 2: A ball is released from rest at a height h above the floor. Consider the system to be the ball and Earth. Air resistance can be ignored.

a) On the following axis, sketch graphs of: U_g vs. distance fallen, K vs. distance fallen, and total mechanical energy vs. distance fallen. Clearly label each line or provide a key.



b) U_g vs. time, K vs. time, and total mechanical energy vs. time. Clearly label each line or provide a key.

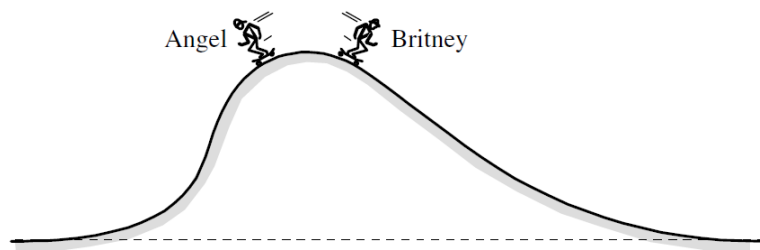


c) When the ball is halfway to the ground (at $1/2 h$), is its kinetic energy more than, less than, or exactly equal to half of its maximum kinetic energy? Explain.

d) When the ball is halfway through falling (at $t = 1/2 T$), is its kinetic energy more than, less than, or exactly equal to half of its maximum kinetic energy? Explain.

TIPERs B4-CT28: SKATEBOARDERS ON A HILL—TIME, SPEED, KINETIC ENERGY, AND WORK

Starting from rest, Angel and Britney skateboard down a hill as shown. Angel rides down the steep side while Britney rides down the shallow side. Angel has more mass than Britney. Assume that friction and air resistance are negligible.



(a) Is the speed at the bottom of the hill

(i) *greater* for Angel, (ii) *greater* for Britney, or (iii) *the same* for both skateboarders? _____

Explain your reasoning.

(b) Is the time it takes to get to the bottom of the hill

(i) *greater* for Angel, (ii) *greater* for Britney, or (iii) *the same* for both skateboarders? _____

Explain your reasoning.

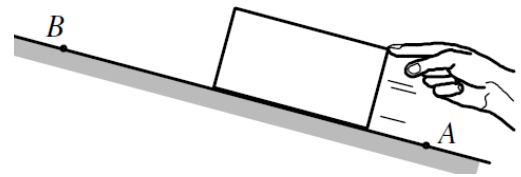
(c) Is the work done by the gravitational force on the skateboarder

(i) *greater* for Angel, (ii) *greater* for Britney, or (iii) *the same* for both skateboarders? _____

Explain your reasoning.

Explain your reasoning.

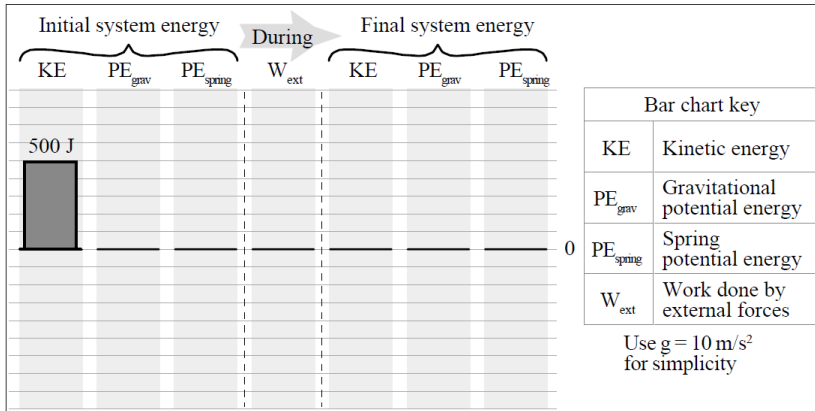
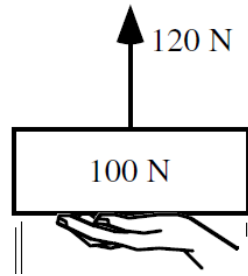
Explain your reasoning.



Explain your reasoning.

TIPERs B4-BCT33: LIFTED BOX MOVING UPWARD II—ENERGY BAR CHART I

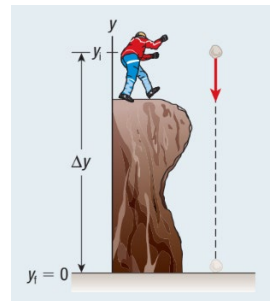
A 100-N box is initially 1.0 m above the ground while moving upward at 10 m/s. A student starts applying a vertical force of 120 N upward with her hand at this point. Complete the energy bar chart for the earth-box system before and after the box has moved upward a distance of 1.0 m. Put the zero point for the gravitational potential energy at the surface of the ground.



Explain your reasoning.

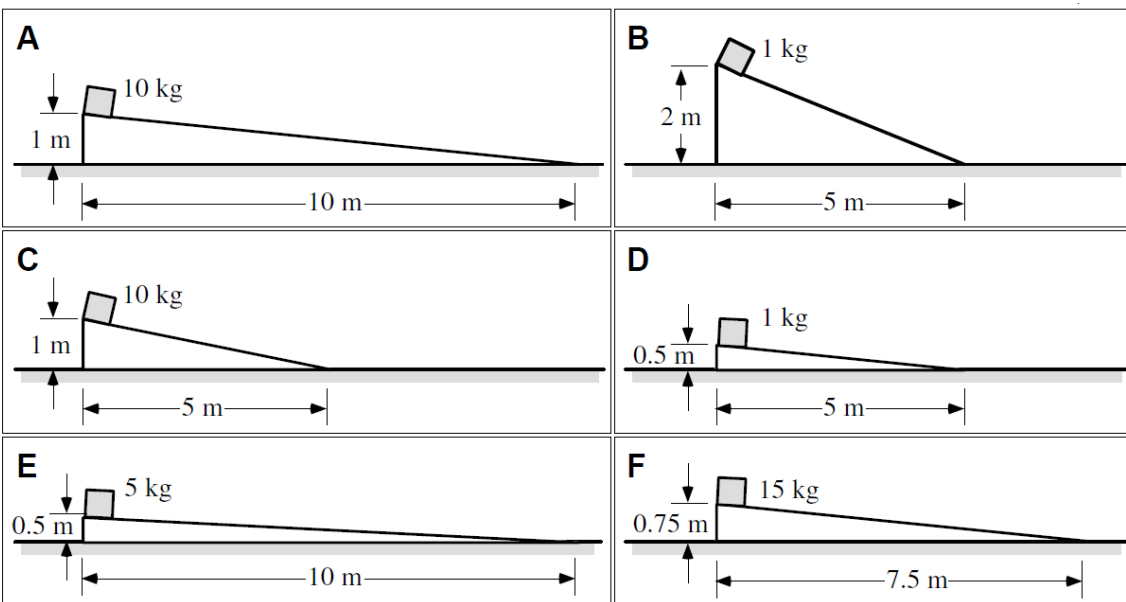
Example 1: Calculating Gravitational Potential Energy

A hiker stands near the edge of a cliff and accidentally drops a rock of mass 1.2 kg to the ground at the base of the cliff 28 m below (figure below). Calculate the potential energy of the rock relative to the ground just before the hiker drops it.



TIPERs B4-RT45: SLIDING MASSES ON INCLINE—KINETIC ENERGY

Shown are blocks that slide down frictionless inclines. All masses start from rest at the top of the incline.



Rank the kinetic energy of the sliding masses the instant they reach the bottom of the incline.

						OR			
1	2	3	4	5	6		All	All	Cannot
Greatest					Least		the same	zero	determine

Explain your reasoning.

Example 2: Applying Gravitational Potential Energy to Determine Mass

A weightlifter raises a loaded barbell 2.2 m. The lift increases the gravitational potential energy of the barbell by 490 J. Determine the mass of the loaded barbell.

Example 3: Gravitational Potential Energy and the Work–Energy Theorem

A physics textbook is 3.6 cm thick and has a mass of 1.6 kg. A student stacks 10 of the books in a single pile on a table. Each book starts from table level before the student lifts it to the top of the stack.

- (a) Calculate the gravitational potential energy of the stack of books with respect to the table.
- (b) Determine the total work done by the student to make the stack of books.
- (c) Calculate the work done by the student to move the books from the stack and set each on the desk.

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding – Lesson 4.3

1. Is gravitational potential energy an absolute value or a relative quantity.
2. When might it be appropriate to select a reference point above the object?
3. State some forms of potential energy.
4. Does mechanical energy refer to a single form of energy?

4.5 The Law of Conservation of Energy

Lesson Goal: Understand the Law of Conservation of Energy and apply it to problem-solving

Success Criteria: At the end of this lesson, I will be able to:

- ✓ To understand that how to define the system matters since only forces exerted by an object or a system external to a system can do work on that system.
- ✓ State the law of conservation of energy
- ✓ Understand that total energy is always conserved, but it is constant only for a closed, isolated system
- ✓ Apply the law of conservation of energy to problem-solving
- ✓ Understand that power is the rate at which energy is transferred into or out of a system or converted within a system
- ✓ Describe power's relationship to work and energy.

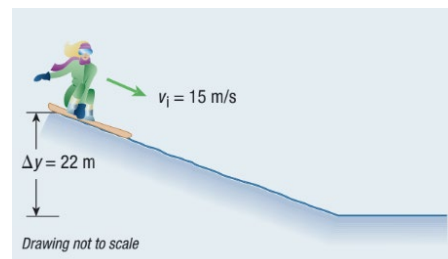
Law of Conservation of Energy

Energy is neither created nor destroyed. It can only change form.

Example 1: Making Connections between Gravitational Potential Energy and Kinetic Energy

A 67 kg snowboarder starts at the top of an icy (frictionless) hill of vertical height 22 m with an initial speed of 15 m/s (figure below).

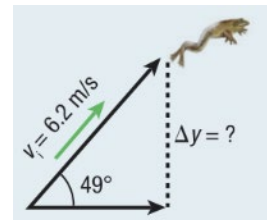
- a) Calculate the snowboarder's mechanical energy at the top of the hill.
- b) Calculate the snowboarder's speed at the midway point and at the bottom of the hill.
- c) Describe the energy transformation that occurs as the snowboarder moves down the hill.



Example 2: Determining Maximum Height and Speed Using the Law of Conservation of Energy

Figure below shows a 0.45 kg bullfrog jumping with an initial speed of 6.2 m/s at an angle of 49° above the horizontal. Assume that energy losses due to air resistance are negligible and can be ignored.

- Calculate the maximum height of the bullfrog's jump.
- Calculate the components of the bullfrog's velocity when it first reaches a height of 0.82 m.



Problem 2 (BFW ch 8 pg336#2): Three balls are thrown off a tall building with the same speed but in different directions. Ball A is thrown in the horizontal direction, ball B starts out at 45° above the horizontal, and ball C begins its flight at 45° below the horizontal. Which of the following both correctly ranks the speed of each ball just before it strikes the ground and best justifies that ranking? Ignore any effects due to air resistance.

- a) $B \geq A \geq C$ because the increase in gravitational potential energy as ball B rises is converted to additional kinetic energy as the ball falls, while the direction in which ball A is thrown does not cause an increase in gravitational potential energy and the direction for ball C causes gravitational potential energy to be lost.
- b) $B \geq A \geq C$ because the time of flight of each ball increases in this order, and the longer a ball is in the air, the more kinetic energy it acquires.
- c) $C \geq A \geq B$ because the kinetic energy in the downward direction is greatest for ball C, zero for ball A, and negative for ball B.
- d) $A = B = C$ because in each case, the ball-Earth system has the same initial kinetic and potential energies and the same final potential energy, so energy conservation requires that they have the same final kinetic energy.

Problem 3 (BFW ch 8 pg336#3): A 1-kg rock is dropped from a cliff 100 m above the sea. The rock strikes the sea and continues to fall through the water until coming to rest on the sea floor. Neglect air resistance but consider drag force as the rock falls through the water. Which of the following best describes the rock–Earth system and its energy as the rock is falling?

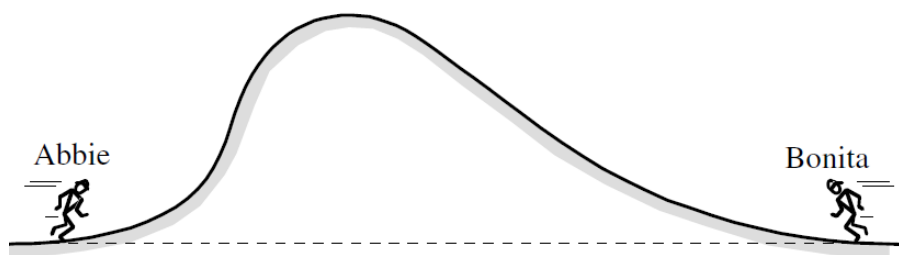
- a) The system is closed and isolated, and all internal forces are conservative, so energy is conserved and mechanical energy is constant.
- b) The system is closed and isolated until the rock strikes the sea surface, so mechanical energy is constant, but as the rock falls through the water the system is no longer isolated and energy is not conserved.
- c) The system is closed and isolated with a constant mechanical energy until the rock strikes the water, when a nonconservative force is exerted on the rock, so mechanical energy is not constant although total energy is conserved.
- d) The system is closed and isolated with a constant mechanical energy until the rock strikes the water, when a nonconservative force is exerted on the rock, so energy is transferred out of the system and total energy is no longer conserved.

Power is the rate of work done by a force over time, or the rate at which the energy of an open system changes.



TIPERs B4-CT48: RACE UP A HILL II—WORK AND POWER

Abbie and Bonita decide to race up a hill that is 30 m high. Abbie takes a path that is 60 m long while Bonita uses a path that is 100 m long. It takes Abbie 40 seconds because her route is steep, while Bonita runs up her path in 30 seconds. They both start from rest at the same height and stop at the top. Abbie has a weight of 700 N, and Bonita has a weight of 500 N.



(a) Is the work that Abbie does in going up the hill (i) *greater than*, (ii) *less than*, or (iii) *the same as* the work that Bonita does in going up the hill? _____

Explain your reasoning.

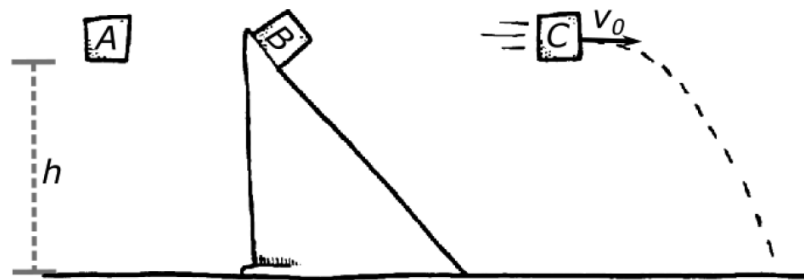
(b) Is the power generated by Abbie in going up the hill *greater than*, *less than*, or *the same as* the power generated by Bonita in going up the hill? _____

Explain your reasoning.

Example 3: Power as a Rate of Change in Kinetic Energy

A car accelerates from rest to a speed of 27.8 m/s in 7.7 s. The mass of the car is 1.1×10^3 kg. Ignoring friction, determine how much power the car requires.

Scenario 1: Three identical blocks are released simultaneously from height h . Block A is dropped from rest, Block B slides from rest down an incline (where friction may be neglected), and Block C is launched horizontally with speed v_0 .



Dominique and Carlos are trying to determine which block will have the fastest speed just before hitting the ground and which block will reach the ground first.

Dominique states, “They will all have the same speed before hitting the ground. Remember that we saw this demonstration in class, and it didn’t matter how the blocks were dropped, they all landed at the same time. The speed will be the same because of the conservation of energy. All blocks have the same gravitational potential energy to start with. This is the total energy and converts into kinetic energy.”

Carlos states, “I think you’re mixing up time and speed. They won’t have the same speed at the end because Block C has an initial velocity. They will all land at the same time, though.”

PART A:

i. Which aspects of Dominique’s reasoning, if any, are correct? Explain your answer.

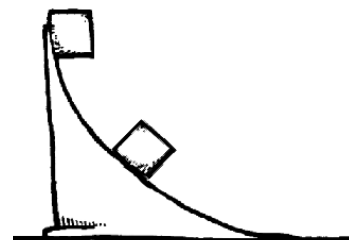
ii. Which aspects of Carlos's reasoning, if any, are correct? Explain your answer.

iii. Which parts of Dominique's reasoning are incorrect? Explain how you know.

iv. Which parts of Carlos's reasoning are incorrect? Explain how you know.

PART B: A fourth identical block is now released from rest at the top of a curved track of radius R . Draw free-body diagrams showing and labeling the forces (not components) exerted on the block at each of the two positions shown at right. Draw the relative lengths of all vectors to reflect the relative magnitudes of all the forces.

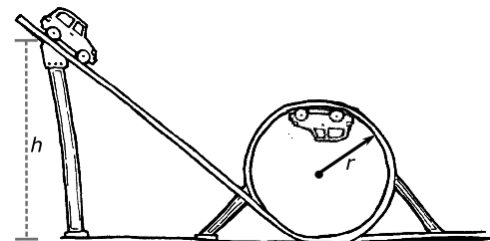
PART C: Blake comes up with the following equation for the work done by the normal force during the slide: $W_{\text{normal}} = \pi/2 * Rmg * \cos\theta$, where R is the radius of curvature of the ramp, and θ is the angle through which the box has traveled.



i. Angela suggests that the equation could be incorrect. Does the equation make physical sense? Briefly explain your reasoning.

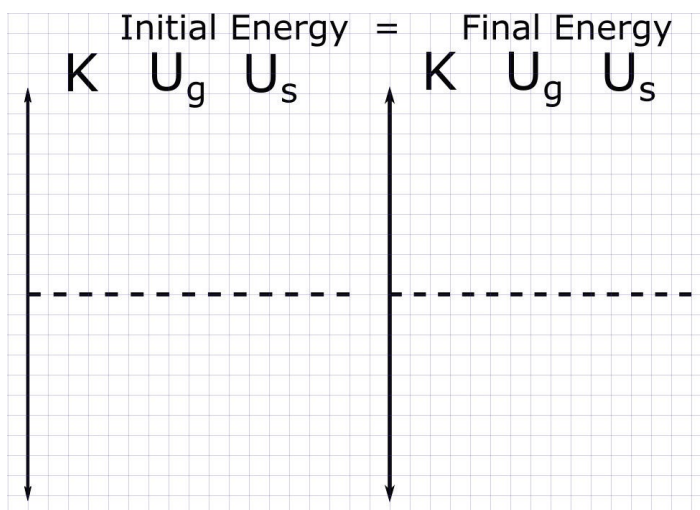
ii. Dominique says that $W_{\text{normal}} = 0$ regardless of the physical situation. Explain why this claim makes physical sense.

Scenario 2: A toy car is released from rest on a smooth track with a loop de loop. The car is released from height h such that it never loses contact with the track. The system includes the car and Earth. Rotational effects from the wheels, friction, and air resistance can be ignored.



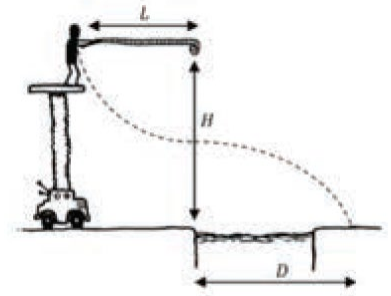
PART A: Draw and label free-body diagrams on the figure shown above that depicts the forces (not components) exerted on the car as it goes down the ramp and at the top of the loop. Draw the relative lengths of all vectors to reflect the relative magnitudes of all the forces. Each force must be represented by a distinct arrow starting on and pointing away from the car.

PART B: Fill in the energy bar chart below with the initial energy when the car is first released from height h energy when the car is at the top of the loop.



PART C: Citing the bar chart, explain why the release height must be greater than the diameter of the loop.

Scenario 3: A designer is creating an obstacle for an obstacle course where a person starts on a moveable platform of height H from the ground. The person grabs a rope of length L and swings downward. At the instant the rope is vertical, the person lets go of the rope and attempts to reach the far side of a water-filled moat. The left side of the moat is directly below the position where the person will let go of the rope. The designer runs several tests in which the rope has different lengths and moves the platform such that the rope is always initially horizontal. The designer notices that the person cannot land on the other side if the length L is very short.



The designer also notices that the person also cannot land on the other side if the length L is very close to the height H .

Assume the size of the person is much smaller than the lengths L and H . Let D represent the horizontal distance from below the release point to where the person lands.

PART A: Answer the following qualitatively, citing physical principles but without manipulating equations.

i. Why does the person land in the moat if the rope's length is very short?

ii. Why does the person land in the moat if the rope's length is nearly the same as the height of the platform?

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding – Lessons 4.5

1. If a ball is tossed upward at the same speed as it is dropped from, will it rebound to the same height it was dropped from?
2. Imagine an object is placed on a ramp. Does the gravitational potential energy of an object depend on the slope of the ramp?
3. Consider the kinetic energy of an object. When is the maximum height achieved
4. Use the law of conservation of energy to analyze the energy conversions as a water balloon falls.

4.6 Elastic Potential Energy and Simple Harmonic Motion

Lesson Goal: Understand the Hooke's Law and elastic potential energy

Success Criteria: At the end of this lesson, I will be able to:

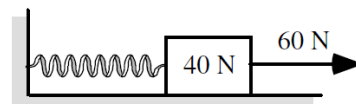
- ✓ Explain the concept of the restoring force in a spring
- ✓ Define simple harmonic motion (SHM)
- ✓ Relate simple harmonic motion to the direction of the restoring force in a spring
- ✓ Define elastic potential energy
- ✓ State Hooke's law and use it in problem solving
- ✓ Recognize that the elastic potential energy of a system is independent of the magnitude of the mass connected to the end of the spring
- ✓ Understand the relation between kinetic energy and elastic potential energy in simple harmonic motion

Elastic potential energy

Now that we went through PP6 let's do some examples:

TIPERs B4-BCT35: BOX ATTACHED TO SPRING—ENERGY BAR CHART

A 40-N box is initially at rest on a smooth (frictionless) horizontal surface. An unstretched spring with spring constant 10 N/m connects the box to the wall. A 60 N force is applied horizontally to the right.



Complete the energy bar chart for the spring-block-earth system as the block moves a distance of 2 m. Label the column heights. Set the zero point for the gravitational potential energy at the center of the block.

Bar chart key	
KE	Kinetic energy
PE _{grav}	Gravitational potential energy
PE _{spring}	Spring potential energy
W _{ext}	Work done by external forces

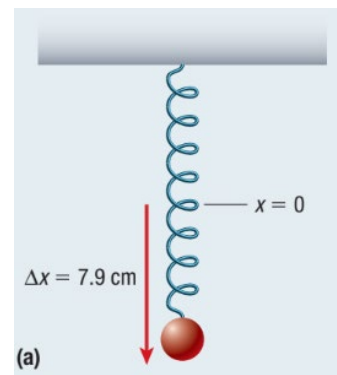
Use $g = 10 \text{ m/s}^2$ for simplicity

Explain your reasoning.

Example 1: Determining and Applying the Spring Constant

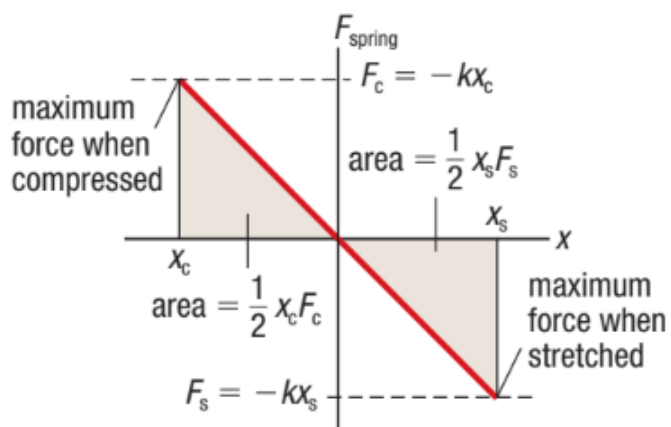
A spring hangs at rest from a support. If you suspend a 0.46 kg mass from the spring, its deflection is 7.9 cm (figure on the right).

- Determine the spring constant.
- Calculate the displacement, in centimetres, of the same spring when a 0.75 kg mass hangs from it instead.
- Suppose the 0.75 kg mass is pushed upward, so that it rises past the spring's unstretched position, compressing the spring. Calculate the net force on the mass when the spring is compressed 5.3 cm. Include a free-body diagram.
- Determine the acceleration of the mass at the position given in (c) once it is released.



Elastic Potential Energy

The graph below shows a graph of the spring force on a mass attached to a spring as the spring is stretched or compressed.

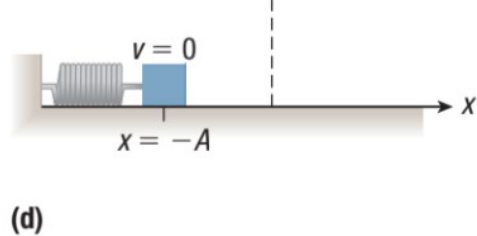
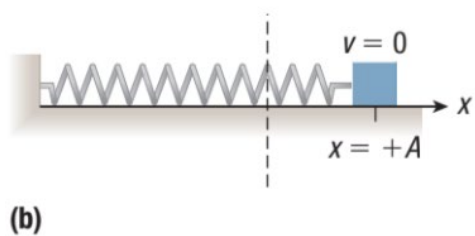
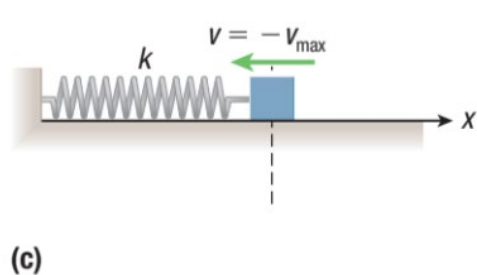
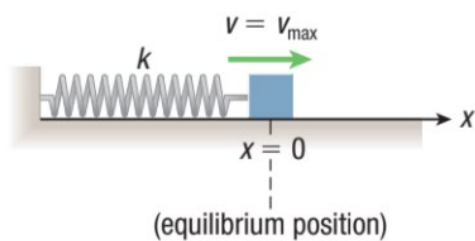


Example 2: Calculate Elastic Potential Energy

A 42 kg teenager balances briefly on a pogo stick, causing the spring in the stick to compress downward by 0.18 m. Determine the elastic potential energy of the teenager.

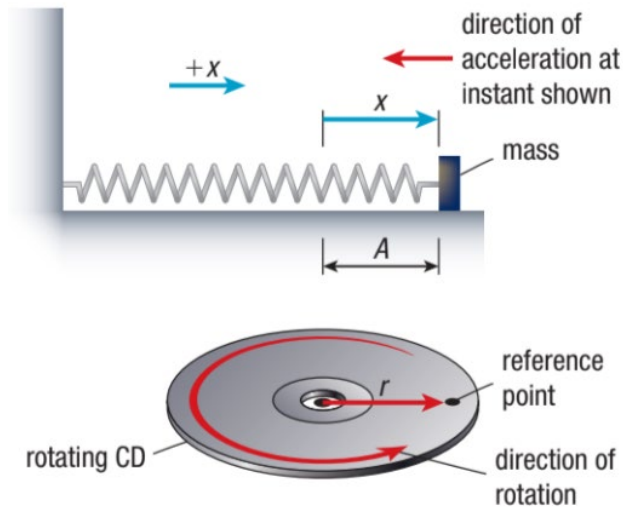
Simple Harmonic Motion

The figure below depicts mass–spring system undergoing simple harmonic motion. Suppose a block is connected to a spring and both are resting on a frictionless surface. The block is at equilibrium when it is resting at its initial position, $x = 0$, as shown in figure (a).



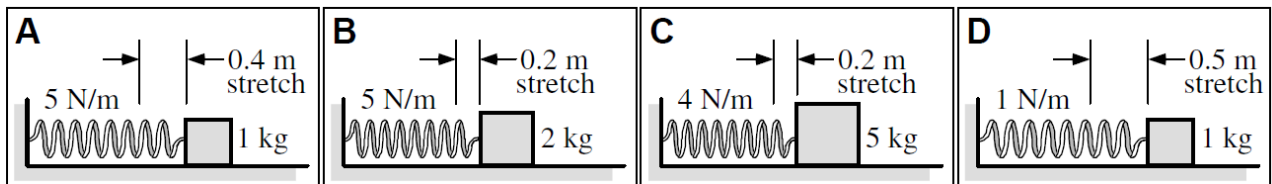
Describing SHM Mathematically

Picture a reference circle, like the CD shown in figure below



TIPERs pg 223 B7-RT01: MASS ON HORIZONTAL SPRING SYSTEMS I—OSCILLATION FREQUENCY

A block rests on a frictionless surface and is attached to the end of a spring. The other end of the spring is attached to a wall. Four block–spring systems are considered. The springs are stretched to the right by the distances shown in the figures and then released from rest. The blocks oscillate back and forth. The mass and force constant of the spring are given for each case.



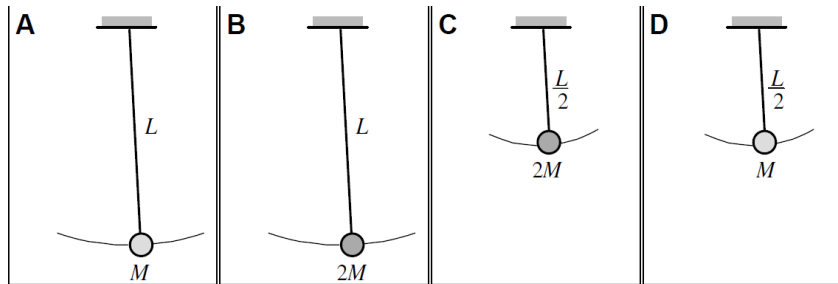
Rank the frequency of the oscillatory motion of the block.

				OR			
1	2	3	4		All	All	Cannot
Greatest			Least		the same	zero	determine

Explain your reasoning.

TIPERs pg 223 B7-RT02: SWINGING SIMPLE PENDULA—OSCILLATION FREQUENCY

The simple pendulum shown in Case A consists of a mass M attached to a massless string of length L . If the mass is pulled to one side a small distance and released, it will swing back and forth. Cases B, C, and D are variations of this system.



Explain your reasoning.

Rank the oscillation frequency of the masses.

				OR		
1	2	3	4		All	Cannot
Greatest			Least		the same	determine

TIPERs B7-QRT05: POSITION-TIME GRAPH OF A CART ATTACHED TO A SPRING—MASS AND PERIOD

A frictionless cart of mass m is attached to a spring with spring constant k . When the cart is displaced from its rest position and released, it

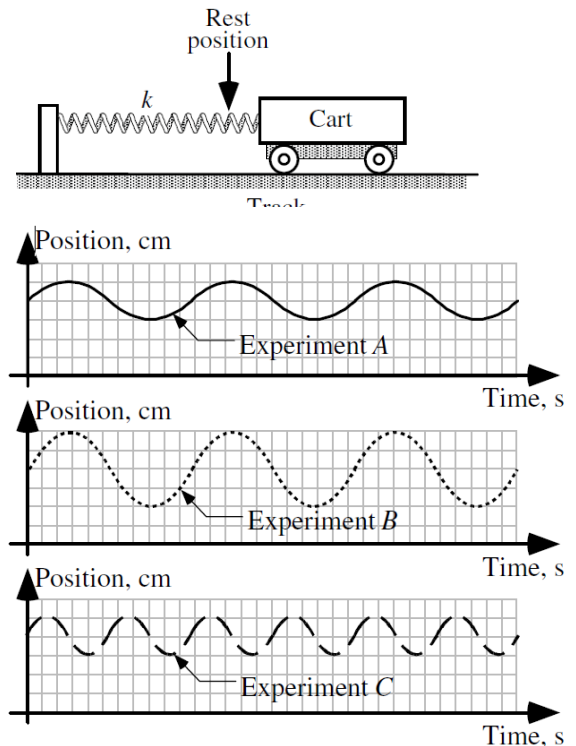
oscillates with a period T that is given by $T = 2\pi\sqrt{\frac{m}{k}}$

The graph of the position of this cart as a function of time is labeled Experiment A. Graphs for two other experiments that use different masses are shown below this. The same spring is used in all three experiments.

(a) Compared to Experiment A, in Experiment B the cart has

- (i) *twice* as much mass.
- (ii) *four times* as much mass.
- (iii) *one-half* the mass.
- (iv) *one-fourth* the mass.
- (v) *the same* mass.

Explain your reasoning.



(b) Compared to Experiment A, in Experiment C the cart has

- (i) *twice* as much mass.
- (ii) *four times* as much mass.
- (iii) *one-half* the mass.
- (iv) *one-fourth* the mass.
- (v) *the same* mass.

Explain your reasoning.

(c) Suppose that in a fourth experiment (Experiment D), the mass used in Experiment A was doubled and the spring was replaced with a spring with spring constant $2k$. The period in Experiment D would be

- (i) *the same* as the period in Experiment A.
- (ii) *double* the period in Experiment A.
- (iii) *four times* the period in Experiment A.
- (iv) *one-half* the period in Experiment A.
- (v) *one-fourth* the period in Experiment A.

Explain your reasoning.

Example 3: Calculating Period and Frequency

When a diving board bends, the restoring force due to the board's elastic properties obeys Hooke's law for an elastic material. Suppose a diver of mass 85 kg stands on a diving board with spring constant 8.1×10^3 N/m. The mass of the board is much smaller than the diver's mass. Calculate the period and frequency at which the board vibrates.

Lesson Reflection

State one thing you learned in this lesson:

*

State one thing you need to go over:

*

Check Your Understanding – Lesson 4.6

1. Give examples from your daily life of devices or tools that use elastic potential energy.
2. Consider a bow and sticky arrow toy that you can purchase from any dollar store. Explain how the energy is transformed.

4.7 Springs and Conservation of Energy

Lesson Goal: Apply the law of conservation of energy in various situations

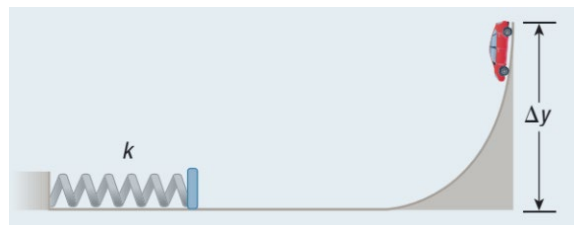
Success Criteria: At the end of this lesson, I will be able to:

- ✓ Apply the law of conservation of energy in various situations
- ✓ Calculate and solve problems involving elastic potential energy
- ✓ Explain why damped harmonic motion is a more accurate model than simple harmonic motion for motions we observe in daily life
- ✓ Explain why it is not possible to create perpetual motion machines

Example 1: Analyzing Energy Transformations

In this problem, you will model a collision and apply conservation of energy to analyze the outcome. A model car of mass 5.0 kg slides down a frictionless ramp into a spring with spring constant $k = 4.9 \text{ kN/m}$ (figure below).

- a) The spring experiences a maximum compression of 22 cm. Determine the height of the initial release point.
- b) Calculate the speed of the model car when the spring has been compressed 15 cm.
- c) Determine the maximum acceleration of the car after it hits the spring.



Example 2: Using Elastic Potential, Kinetic, and Gravitational Potential Energies

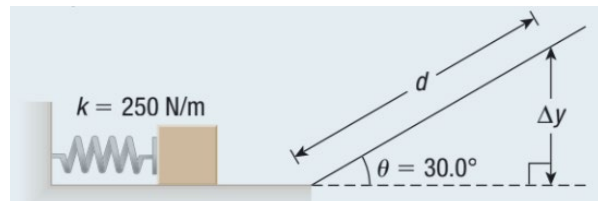
A 48 kg child bounces on a pogo stick. At the lowest point of one bounce, the compressed spring in the stick has 120 J of elastic potential energy as it compresses 0.19 m. Assume that the pogo stick is light enough that we can ignore its mass.

- a) Determine the child's maximum height during the jump following the bounce.
- b) Determine the child's maximum speed during the jump.

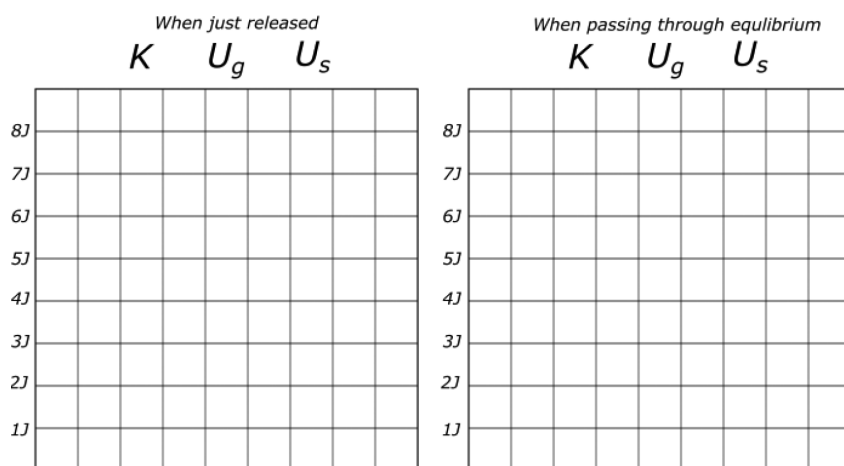
Example 3: A Block Pushed Up a Frictionless Ramp by a Spring

A block with a mass of 2.0 kg is held against a spring with spring constant 250 N/m. The block compresses the spring 22 cm from its equilibrium position. After the block is released, it travels along a frictionless surface and then up a frictionless ramp. The ramp's angle of inclination is 30.0° , as shown in figure below.

- Determine the elastic potential energy stored in the spring before the mass is released.
- Calculate the speed of the block as it travels along the horizontal surface.
- Determine how far along the ramp the block will travel before it stops.



Dominique says that the height above the ground doesn't matter; only the change in height is relevant. She considers the equilibrium position to be zero gravitational potential energy. Complete the energy bar charts for the block-Earth-spring system when the block is initially released.



Block/Earth/Spring system when $U_g = 0$ at equilibrium

PART D: Use the results of your bar charts to support Dominique's claim that the zero point of gravitational potential energy does not affect the behavior of the block-Earth-spring system. Reference specific details of the students' energy bar charts in your answer.

Problem 1 (BFW): A block of mass m is sliding along a horizontal surface with negligible friction. The speed of the block is v_0 . The block collides with and sticks to a horizontal spring with spring constant k and compresses the spring a distance x_1 before coming momentarily to rest. A second block of mass $2m$ is then launched toward the same spring with the same initial speed. What is the maximum compression of the spring in this second case, in terms of x_1 ?

- A) $2x_1$
- B) $\sqrt{2} x_1$
- C) x_1
- D) $4x_1$

Lesson Reflection

State one thing you learned in this lesson:

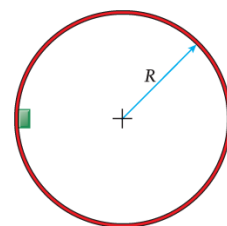
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State one thing you need to go over:

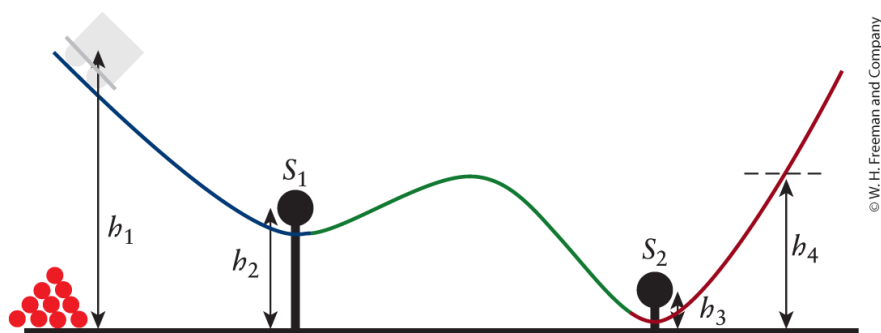
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Review Lesson - Work and Energy Problem Solving

1. (pg 372 #16) A small block of mass M is placed halfway up on the inside of a circular loop of radius R , as shown in the figure. The size of the block is very small compared to the radius of the loop. Determine an expression for the minimum downward speed with which the block must be launched in order to guarantee that it will make the full loop.

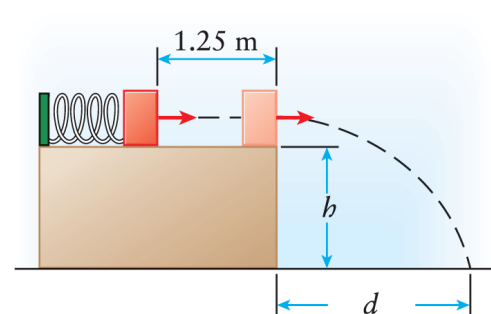


2. (pg. 372 #20) A team of students is building a model roller coaster. They find that when the cart is released from the height h_1 , as shown in the diagram, the cart stops rolling at height h_4 . Sofia claims that it is a drag force that is responsible for the failure of the cart to maintain a constant mechanical energy. Jackie claims that it is friction between the cart and the track or friction between the axles and the wheels. The track is composed of three sections (blue, green, and red in the diagram). Each section is the same length, l . They have placed sensors at positions S_1 and S_2 that can be used to measure the speed of the cart at these points. The body of the cart is an open box into which they can place small metal spheres. The box can be removed, leaving just the wheels, and there is a second shorter box that can be mounted on the wheels.



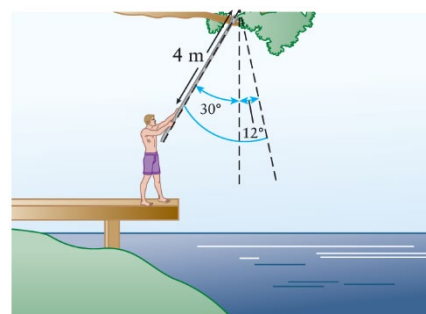
- Express Sofia's claim in terms of mechanical energy and the drag force on the cart and identify the possible independent and dependent variables in this expression.
- Express Jackie's claim in terms of mechanical energy and the friction force on the cart and identify the possible independent and dependent variables in this expression.
- Describe two different designs that Sofia can use to collect the evidence needed to test her claim.
- Describe two different designs that Jackie can use to collect the evidence needed to test her claim.

3. (pg. 372 #21) An object with mass m is placed against the free end of a spring (with spring constant k) that is compressed a distance Δx from the equilibrium length, which is a distance of 1.25 m from the edge of the tabletop. Once released, the object slides across the tabletop and eventually lands a distance d from the edge of the table onto the floor, as shown.



- Define your system and construct a diagram for the analysis, including the labelling of variables.
- Express the velocity of the block at the instant that it leaves the table in terms of k , m , and Δx .
- Express the velocity of the block at the instant that it leaves the table in terms of d and h .
- Define values of system variables as follows: $d = 1.60$ m; $h = 1.0$ m; $m = 0.10$ kg; $k = 75$ N/m; and $\Delta x = 0.15$ m. Justify the claim that the table exerts a friction force on the block.
- Calculate the coefficient of kinetic friction for the interaction of the tabletop and the block.

4. (pg. 373 #22) Logan, with mass $m = 75$ kg, is standing at the edge of a dock holding onto a rope, which is suspended from a tree branch above. The distance along the rope from Logan's hands to the point of attachment to the tree is $L = 4.0$ m. The rope is taut and makes an angle, θ , of 30° with the vertical direction. Logan very gently steps off of the dock and swings in a circular arc until he releases the rope when it makes an angle, ϕ , of 12° with the vertical, but on the other side of the branch.

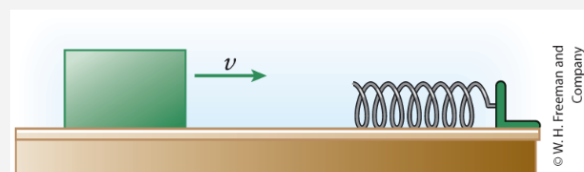


- Define your system so that you may use it to complete the rest of this problem and construct a diagram showing the initial (i) and final (f) configurations of Logan's swing.
- Apply the principle of energy conservation to the system in terms of K_i , K_f , U_i , and U_f in terms of θ , ϕ , L , g , m , v_i , and v_f .
- Evaluate the expression in part (b) to determine Logan's speed when he releases the rope.
- Justify the selection of additional data required to determine the horizontal displacement of Logan from the edge of the dock to when he strikes the water and summarize the mathematical model from which the displacement can be calculated.

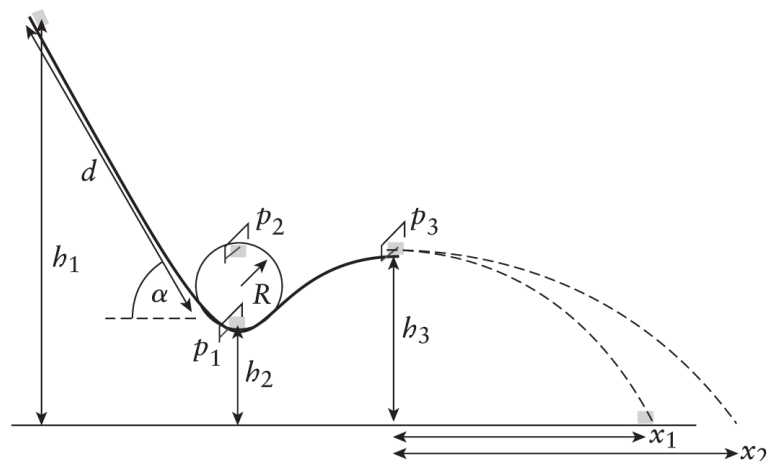
5. (pg. 373 #23) The unknown spring constant of a spring is determined by measurement of the maximum compression, x , when the spring is struck by a 100-g block with a known velocity. The surface over which the block travels exerts a negligible friction force on the block. The data obtained from the measurements for a particular spring are shown in the table.

- Express the compression of the spring in terms of the velocity of the block using the principle of energy conservation.
- Analyze the data to determine the spring constant in units of N/m.

v_i (m/s)	x (cm)
5	4.5
10	9
15	14
20	18
25	22



6. (pg. 373 #24) In the apparatus shown, a block with mass m is released from height h_1 , descends along a track with an inclination α , does a loop with radius R and minimum height h_2 , and is launched horizontally from a height $h_3 = h_2 + 2R$. The team of students who designed and built the track predicted x_2 as the distance from the launch position to landing. When they tested the prediction they observed the distance x_1 . They knew that there were friction forces because they heard the block sliding, so they want to add a frictional work term to their energy conservation analysis. They have placed three photogates at positions at the top and bottom of the circular loop in the track and at the point where the block leaves the track.



- First apply energy conservation for a closed system to obtain their prediction x_2 . The team believed that most of the frictional work occurs on the descent before the loop, given that $d > R$.
- Apply energy conservation to obtain the relationship that the team used to measure this work in terms of the speed measured at photogate p_1 .
- Explain why the team predicted that the speeds at photogates p_2 and p_3 would be the same.

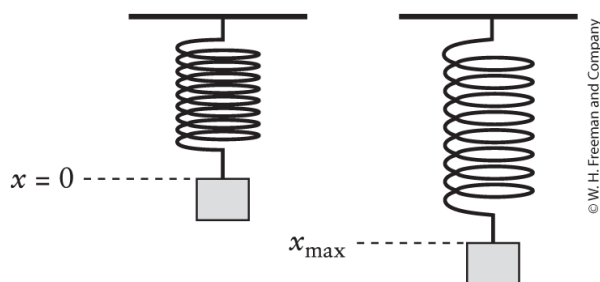
In fact, the team discovered that the speeds at photogates p_2 and p_3 were significantly different. So, they decided that if they were going to succeed in understanding the system they would need to estimate the coefficient of kinetic friction.

d) Apply energy conservation and Newton's second law to express the friction coefficient, μk , in terms of g , d , α , and the velocity at photogate $p1$, $v1$, before the block passes through the loop.

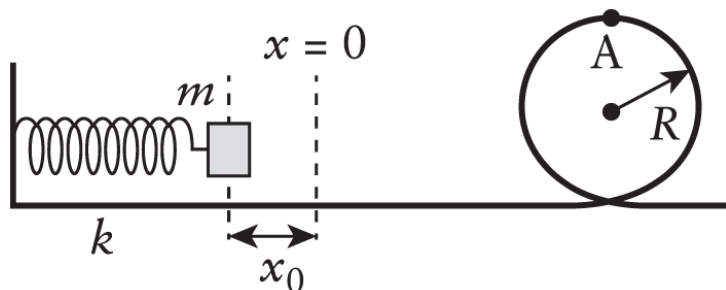
e) The team then assumed that the frictional work could be neglected on the shorter lengths of the track as the block slides through the loop and up to the launch position. To test this claim, the team modified the prediction for x_2 in part (a) to include the friction force along the track before the loop. Obtain their expression.

f) Is there a problem with their assumption in part (e)?

7. (pg. 374 #27) A block of mass m is connected to a spring of unknown constant k that is hung vertically from a support, as shown. With the spring initially unstretched, the block is released from rest and descends, stretching the spring a distance $x_{\max}$ before coming momentarily to rest and moving back upward. Derive an expression for k in terms of m , $x_{\max}$, and any relevant physical constants.



8. (pg. 374 #28) A block of mass m is compressed a distance x_0 against a spring of constant k and released from rest. The block is not connected to the spring. The block then slides with negligible friction along a track and around a vertical circle of radius R , as shown. What is the minimum compression, x_0 , such that the block will slide completely around the loop without losing contact at point A?

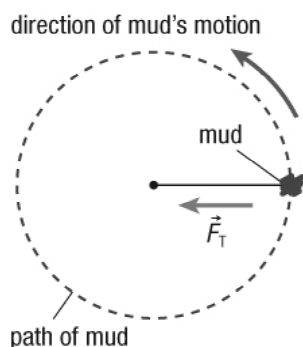


Chapter 4 - Check Your Knowledge – Definitions

Term	Definition
___ work	1. any spring that obeys Hooke's law; it does not experience any internal or external friction
___ joule	2. stored energy an object has because of its position and the applied gravitation force
___ kinetic energy	3. the potential energy due to stretching or compressing of an elastic material
___ work-energy theorem	4. the constant of variation between the force exerted by an ideal spring and the spring's displacement
___ potential energy	5. the maximum displacement of a wave
___ mechanical energy	6. the energy an object has because of its motion
___ gravitational potential energy	7. a machine that can operate forever without restarting or refuelling
___ law of conservation energy	8. periodic motion in which the acceleration of the moving object is proportional to its displacement
___ power	9. the amount of force exerted by a string is directly proportional to the displacement of the string
___ biochemical energy	10. the total work done on an object equals the change in its kinetic energy
___ spring constant	11. Energy is neither created nor destroyed. It can only change form.
___ ideal spring	12. stored energy an object has that can be converted into another form of energy
___ elastic potential energy	13. the sum of an object's kinetic and potential energies
___ Hooke's Law	14. the product of the magnitude of an object's displacement and the component of the applied force in the direction of the displacement
___ amplitude	15. the rate of work done by a force over time; or the rate at which the energy of an open system changes
___ simple harmonic motion	16. periodic motion affected by friction
___ perpetual motion machine	17. a type of chemical potential energy stored in the cells and other basic structures of biological organisms
___ damped harmonic motion	18. the SI unit of work and energy; a force of 1N acting over a displacement of 1 m does 1 J of work

Chapter 4 –Communication Questions

1. A patch of mud has stuck to the surface of a bicycle tire as shown. The stickiness of the mud is the centripetal or tension force that keeps the mud on the tire as it spins. Calculate the work done on the mud as the tire makes one revolution, if the mud stays on the tire.



2. Most of the examples in this chapter did not consider the presence of friction or else assumed it to be negligible. Do you think this is realistic?
3. A spring stores potential energy when it is stretched or compressed. Are there any other simple common devices that behave in this way to store potential energy?
4. List some disadvantages of hydroelectric power generation.
5. Hydro is the largest source of energy generation in Ontario. Ontarians view this as a clean source of energy. Do you agree that we should develop more hydroelectric power generation?
6. List some advantages of hydroelectric power generation.